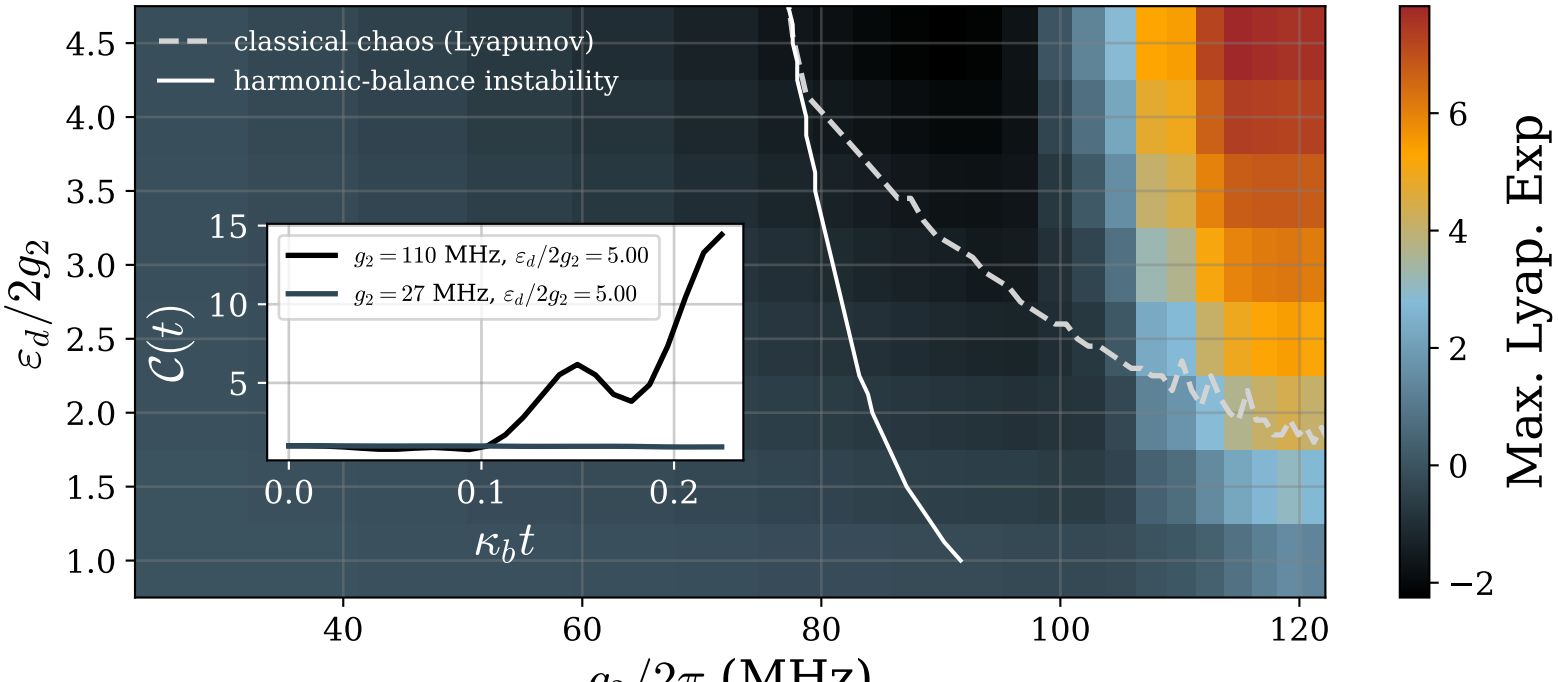
classical chaos (Lyapunov)
harmonic-balance instability
$\varepsilon_d/2g_2$
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
40
60
80
100
120
$g_2/2\pi$ (MHz)
$g_2 = 110$ MHz, $\varepsilon_d/2g_2 = 5.00$
$g_2 = 27$ MHz, $\varepsilon_d/2g_2 = 5.00$
$\mathcal{C}(t)$
15
10
5
0.0
0.1
0.2
$\kappa_b t$
Max. Lyap. Exp
6
4
2
0
−2